Null Choir

Reverse Engineering



I began by triaging the provided files: stripped ELF64 binary `null_choir`, bytecode stream `canticle.score` (magic NCS2), and trace capture `echoes.trace` (magic ECH8).

Reverse engineering in IDA Free revealed a custom stack-and-register bytecode virtual machine interpreting 4,096 instructions with 16 distinct opcodes operating over a 48-byte key buffer.

The VM enforced intermediate state checks validated against authenticated execution snapshots in echoes.trace. I implemented an emulator in Python to symbolically invert each arithmetic transformation block backward to recover the 48-byte input.

Finally, I tested the recovered flag against the executable / service:

`./null_choir 'SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}'`

VM Verification: Harmony.

Flag Captured

`SecLeaf{th3_ch01r_s1ng5_1n_5h4d0w_m4ch1n35!!!!!}`

BY:JOKER_53